

Edwin Rojas-Torres

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EDUCATION

University of California, Santa Cruz Sep. 2019 - Dec. 2023

B.S. Computer Engineering

Relevant Coursework: Data Structures, RTOS, Computer Architecture, Embedded Systems, Distributed Systems

SKILLS

Hardware Design & FPGA: RTL Design (SystemVerilog), Xilinx, PYNQ, Xilinx Runtime (XRT)

Embedded & Automotive Systems: Embedded C/C++, CAN Bus/CAN FD, LIN Bus, Vector CANoe

Software Engineering: Python, Node.js, PostgreSQL, MongoDB, Vite, Angular, React, Jest

Languages: English (Native), Spanish (Native)

Systems & Tools: Linux, Bash, Git, GitHub, Docker, CI/CD, AWS, RESTful APIs, Claude, Agentic Services

PROFESSIONAL EXPERIENCE

Fullstack Engineering Intern

Sep. 2025 - April. 2026

Blue Trail Software, San Francisco, CA

- Eliminated a production API overload by redesigning the Contentful data-fetching layer, cutting API calls from ~3 million to ~12,000 per day (99.6% reduction). Diagnosed and shipped the same business day.
- Architected OAuth 2.0 single-sign-on infrastructure using the Google OAuth API and AWS Secrets Manager, enabling secure access for ~150 internal users with zero credential-exposure incidents.
- Implemented an i18n localization framework supporting four languages (EN/ES/FR/DE), increasing accessible surface area of the application by 4x.
- Collaborated cross-functionally with design and product to ship UI features on tight sprint cycles using React, Node.js, and REST APIs.

Automotive Engineer (SWAT/OPS)

Apr. 2024 - Oct. 2024

Stellantis, Auburn Hills, MI

- Performed CAN/LIN bus diagnostics using CANoe on production automotive ECUs, reducing diagnostic turnaround time for wiring harness validation across Michigan assembly plants.
- Executed root-cause analysis on live vehicle network faults, isolating signal integrity issues in CAN frames that were causing intermittent ECU failures on the assembly line.
- Validated wiring harness continuity and pin-out integrity against OEM schematics, catching assembly defects before downstream integration.

RESEARCH

DRAM Leakage Characterization via FPGA-Accelerated DMA

Jan. 2023 - Dec. 2023

Neuromorphic Computing Group, Santa Cruz, CA

- Modified DMA Engine IP in RTL (Vivado) on a Xilinx Alveo U200 to expose fine-grained control over DRAM access patterns, enabling Rowhammer-style leakage timing at nanosecond resolution.
- Developed low-level firmware in Python via PYNQ and Xilinx XRT to orchestrate DMA transactions directly against FPGA-attached DRAM and the Samsung SmartSSD NAND interface.
- Reverse-engineered and bypassed vendor safety locks on the Samsung SmartSSD after the Alveo's on-board DRAM proved uncontrollable under standard XRT, enabling direct NAND access.
- Built a Python-based automated test framework achieving consistent ~0.02 s DRAM leakage timing across millions of measurement cycles; thesis work subsequently extended by UC Santa Cruz researchers.